

RESTAURANT TABLE RESERVATION SYSTEM REPORT

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ABSTRACT

This report describes the design and implementation of a Restaurant Table Reservation System using PHP and MySQL. The system enables customers to make online reservations by selecting a date, time slot, and party size, while ensuring that only available tables are assigned. It utilizes a structured relational database with entities for customers, reservations, tables, and time slots.

An availability-checking algorithm is implemented to prevent double-booking by filtering out tables already reserved for a given time. Data validation and prepared statements are used to maintain data integrity and enhance security. Transaction management further ensures consistency during concurrent booking operations.

The system also includes session-based state management for temporary reservation holding and an administrative interface for managing bookings. Overall, the application provides an efficient and secure solution for managing restaurant reservations.

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1. TECHNICAL REPORT

1.1 Introduction

The Restaurant Table Reservation System is a web-based application developed using PHP and MySQL. It allows customers to reserve tables online based on availability, while preventing overbooking and ensuring efficient table management.

1.2 Algorithm for Checking Table Availability

The system determines table availability using a filtering algorithm based on table capacity and existing reservations.

Steps:

1. User inputs date, time, and party size.
2. System retrieves all tables where capacity is greater than or equal to the party size.
3. System retrieves all tables already reserved for the selected date and time.
4. Reserved tables are excluded from the available tables list.
5. Remaining tables are displayed to the user.

Pseudocode:

```
INPUT date, time, party_size
```

```
available_tables = SELECT tables WHERE capacity >= party_size  
reserved_tables = SELECT table_number FROM reservations WHERE date  
AND time
```

```
available_tables = available_tables - reserved_tables
```

```
DISPLAY available_tables
```

This ensures that only suitable and unreserved tables are shown to the user.

1.3 Database Locking Considerations

To prevent multiple users from booking the same table simultaneously, the system uses database transactions.

Approach:

- A transaction begins before inserting a reservation.
- The system checks if the selected table is still available.
- If available, the reservation is inserted and the transaction is committed.
- If not, the transaction is rolled back.

Benefits:

- Prevents race conditions.
- Ensures data consistency.

- Avoids duplicate bookings.

1.4 Time Slot Management Logic

The system uses a dedicated table to define valid booking times.

Key Features:

- Only predefined time slots (e.g., 17:00, 18:00, 19:00) are allowed.
- Users select time slots from a dropdown menu populated from the database.
- Each reservation corresponds to one time slot.

Advantages:

- Simplifies availability checking.
- Prevents invalid or inconsistent time entries.
- Ensures uniform booking intervals.

1.5 Prevention of Overbooking

Overbooking is prevented through multiple mechanisms:

1. Capacity Filtering: Only tables that can accommodate the party size are considered.
2. Reservation Check: The system verifies that no reservation exists for the same table, date, and time.
3. Transaction Control: Database transactions ensure that simultaneous booking attempts do not result in conflicts.
4. Session-Based Holding: When a user selects a table, it is temporarily held during the payment process.

These measures ensure that each table can only be reserved once per time slot.

2. CONCLUSION

The Restaurant Table Reservation System provides an efficient and secure way to manage table bookings. By combining structured database design, validation mechanisms, and transaction handling, the system ensures reliability and prevents overbooking. Additional features such as POS integration and mobile responsiveness further enhance its practical usability in real-world restaurant environments.